

Task - Learn JS concepts with the help of AI

[Claude prompt log](#)

Changes through the 5 prompts -

ROLE

When I told the model it's a senior JS mentor, it stopped sounding like a textbook and starts sounding like an actual person explaining things to you.

CONTEXT

it doesn't over-explain things you already know or under-explain things you don't. Since I said I learn through analogies, it leaned on real world comparisons like locker in a room instead of jumping straight into jargon.

FEW SHOT EXAMPLES

Instead of describing the format I wanted in words, I just showed it two examples of it. The model picks up the pattern from seeing it done correctly, which worked far more reliably than trying to explain formatting rules in plain English.

CONSTRAINTS

I gave it constraints like keep the analogy grounded, don't write a massive code block. Without constraints, the model tends to drift over a long conversation, so this keeps every answer consistent no matter how far the chat goes. This also helps in keeping the conversation clear and concise.

STEP DECOMPOSITION

This forced the model to slow down and trace through the code line by line before jumping to a final answer. It stops it from skipping steps or hand-waving the logic, which matters a lot when the whole point is understanding, not just getting the right answer. This is because I learnt that models can NOT backtrack in their context window if we give them a generic prompt. When we break down into steps, they can backtrack and pull in tokens from the last request to verify their work.

Comparison of GPT and Claude-

Tone

- **Response 1 (Claude):** It was friendly and conversational. It reads like a helpful tutor using narrative language to explain the concept.
- **Response 2 (ChatGPT):** it was direct and analytical. It uses short statements

Structure

- **Response 1 (Claude):** Narrative paragraphs
- **Response 2 (ChatGPT):** Nothing special. Almost same as Claude.

Accuracy and Consistency

- **Response 1 (Claude):** it was very accurate. The code example stays exactly the same from start to the finish
- **Response 2 (ChatGPT):** High accuracy but I think its answer was more accurate and I understood better with GPT actually.

Analogy

- **Response 1 (Claude):** An example using a backpack. I think this was a great analogy and helped me to grasp the concept at first sight.
- **Response 2 (ChatGPT):** gave me an example using a locker. I think this analogy was much simpler than what Claude sent about the backpack

Final Template that anyone can use to learn concepts in JS step by step, sure-shot:

ROLE: You are a senior JavaScript engineer who has mentored many beginners into professional frontend developers.

CONTEXT: I am a second-year engineering student. I know HTML and CSS well but am new to JavaScript. I learn best through analogies and visual/concrete examples rather than formal definitions.

TASK: Explain the concept of [CONCEPT] in JavaScript.

FEW SHOT EXAMPLES:

Example 1:

Input: What is a variable?

Output:

Definition: A variable stores data.

Example: let age = 20;

Real-life analogy: A labeled box that holds something.

Example 2:

Input: What is a function?

Output:

Definition: A function is reusable code.

Example: `function greet(){}`

Real-life analogy: A coffee machine that gives coffee whenever you press a button.

CONSTRAINTS:

- Use exactly this format: Definition / Example / Real-life analogy
- Keep the analogy concrete, not abstract
- Code example must be short (under 10 lines) and runnable

STEP DECOMPOSITION: Before giving the final Definition/Example/Analogy answer, walk through the code step by step:

1. Define what each key part of the code does individually
2. Trace what happens when it executes, in order
3. Only then synthesize into the final Definition/Example/Analogy format